
Information Retrieval

Readings for this lecture

Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze, Introduction to Information Retrieval, Cambridge University Press. 2008.

<http://nlp.stanford.edu/IR-book/>

What is Information Retrieval?

- The study of finding needed information. i.e., to help users find information that matches their information needs.
 - ✓ Expressed as **queries**
- To be precise, IR is about document retrieval, emphasizing the document as the basic unit.
 - ✓ **Finding documents** relevant to user queries

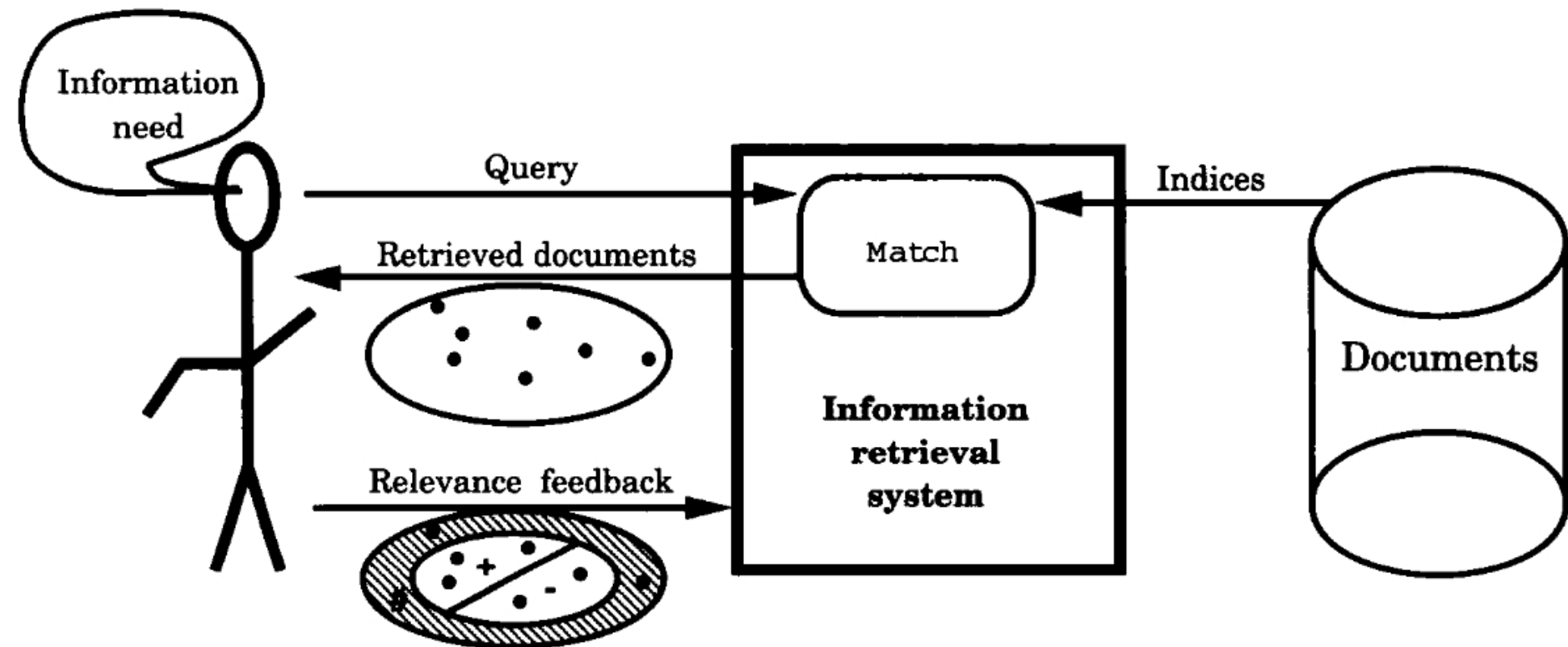
What are the key challenges for Information Retrieval?

- acquisition
- organization
- storage
- retrieval
- distribution of information

Why is Information Retrieval Required?

- Web Search
- E-mail search
- Searching your laptop
- Searching corporate knowledge bases
- Legal information retrieval
-

IR Cycle



Basic Concepts

- **Information Need**: State (of a person) of requiring information for solving an actual problem (e.g.: get rid of mice without hurting them).
- **Query**: Textual representation of the information need, e.g. as entered into a search engine
- **Relevance**: Property of a document with respect to a particular information need. The task of IR is to retrieve relevant documents
- **Relevance Feedback**: Information about the actual relevance of retrieved documents given by the user of an IR system to the system
- **Evaluation**: Measuring the quality of an IR system's performance

Two Major Challenges

- How are document and query represented ?
- How is the relevance of a document to a user query defined?

What is Relevance?

- Given a particular information need, documents in a document collection differ with respect to being relevant or irrelevant
~ “*useful*”, ~ “*interesting*”, ~ “*satisfying*”, ...
 - ✓ **Minimal requirement**: Documents are somehow 'about' the topic
- In practice, the distinction is not necessarily clear-cut:
 - ✓ Information needs (intent) and document contents are often subject to interpretation.
 - ✓ What one user finds relevant, another might find off-topic!
- For any given pair of information need and document, **there is no definitive answer to the question of relevance**

Term-Document Representation: Incidence Matrix

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	1	1	0	0	0	1
Brutus	1	1	0	1	0	0
Caesar	1	1	0	1	1	1
Calpurnia	0	1	0	0	0	0
Cleopatra	1	0	0	0	0	0
mercy	1	0	1	1	1	1
worser	1	0	1	1	1	0

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1 if play contains word,
0 otherwise

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Doc vector "Othello"

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To answer query: take the term vectors for Brutus, Caesar and Calpurnia (complemented) with bitwise AND:

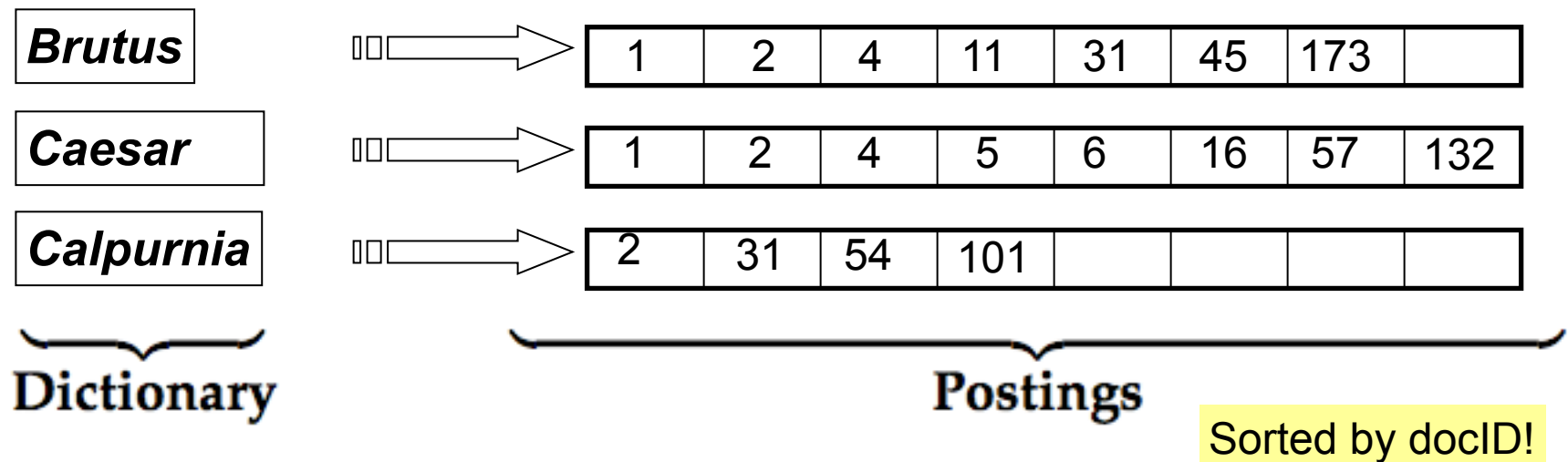
110100 AND 110111 AND 101111 = 100100.

Issues with Incidence Matrix

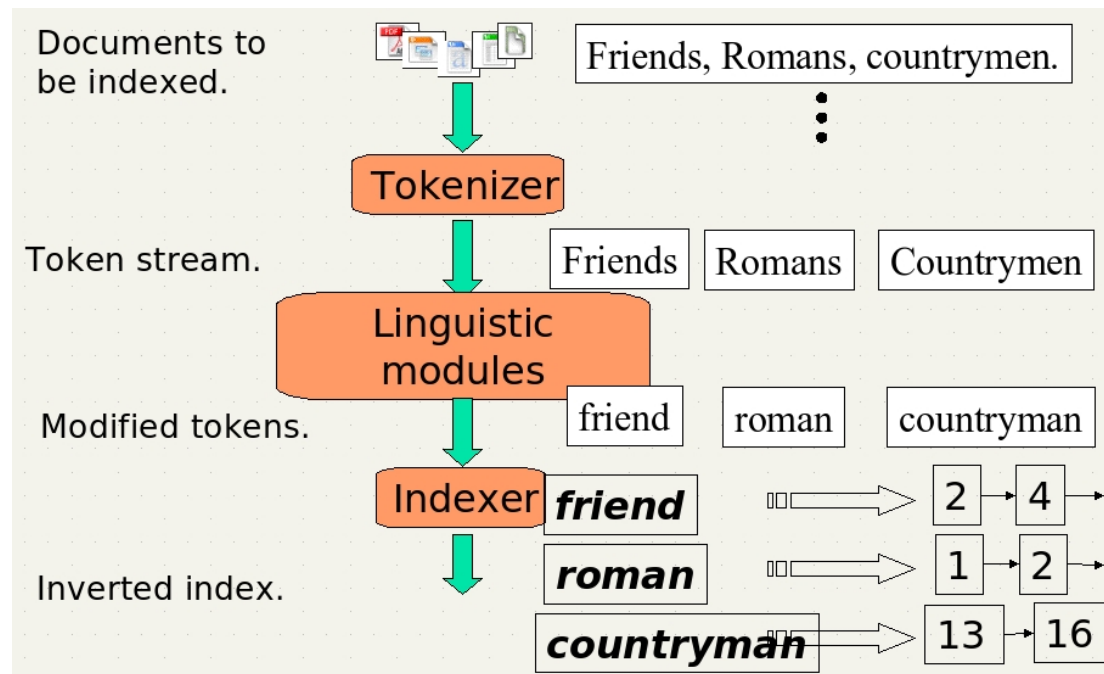
- Consider $N = 1$ million documents, each with about 1000 words.
- Avg. 6 bytes/word including spaces/punctuation
 - 6GB of data in the documents.
- Say there are $M = 500K$ **distinct** terms among these.
- $500K \times 1M$ matrix has half-a-trillion 0's and 1's.
- But it has no more than one billion 1's.: Matrix is sparse.
- Better Idea of representation: we only record 1-positions.

Term-Document Representation: Inverted Index

- For each term t , we must store a list of all documents that contain t .
 - Identify each by a **docID**, a document serial number
 - A posting contains the docID, possibly more information such as position, term frequency, etc.



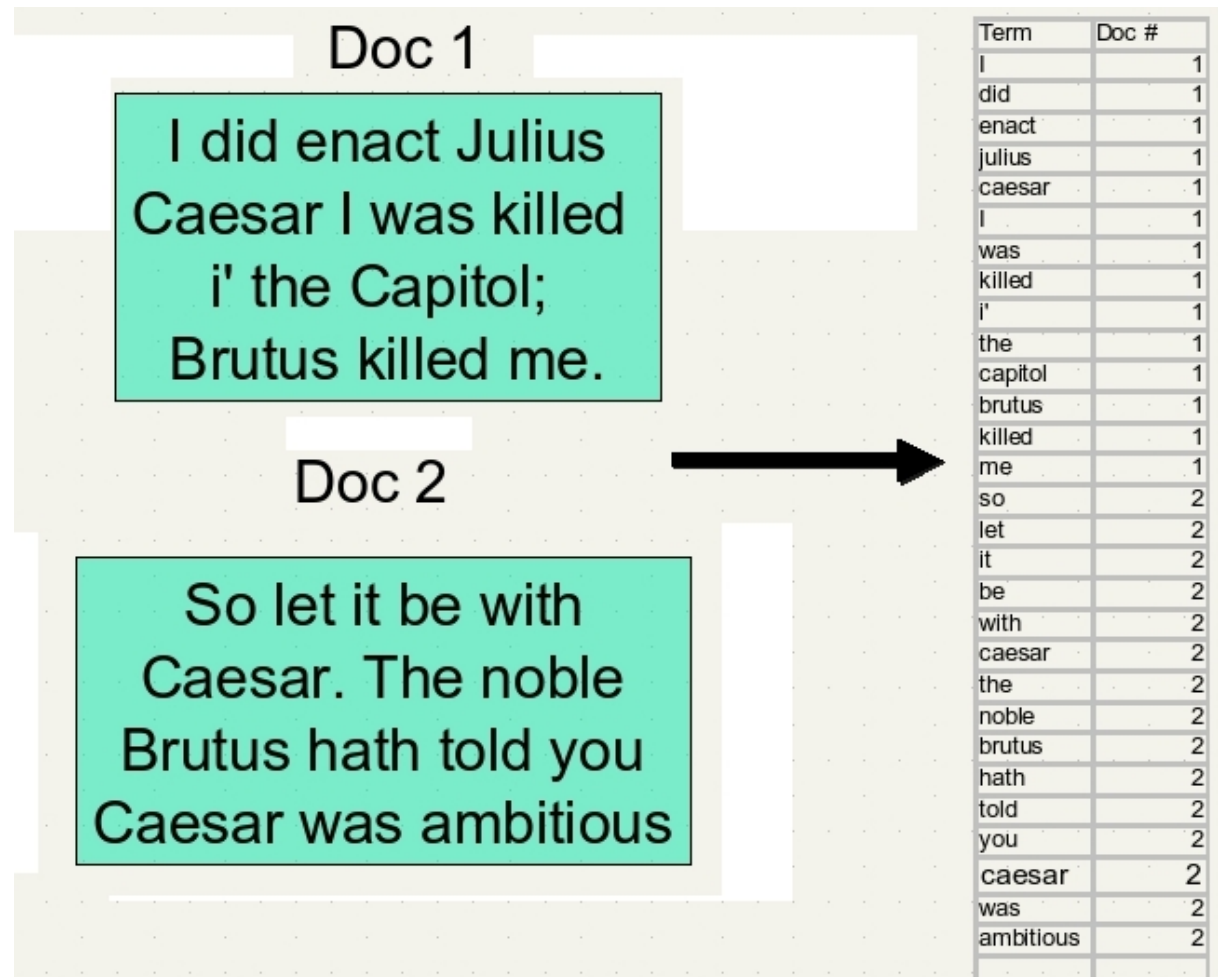
Overview of Index Creation model



- **Tokenizer**: identifies terms in documents
- **'Linguistic' Modules**
 - case folding
 - stemming
 - normalization of dates, numbers etc.
- **Indexer**
 - creates inverted index

Index Creation: Step 1

- The definition of “token” is underlying in the whole system
- Need the same definition of token for all documents and queries



Index Creation: Step 2

- Convert each doc into list of tokens
- Sort by terms
- Merge multiple term entries
- Add frequency information

Term	Doc #		Term	Doc #	Term freq
ambitious	2		ambitious	2	1
be	2		be	2	1
brutus	1		brutus	1	1
brutus	2		brutus	2	1
capitol	1		capitol	1	1
caesar	1		caesar	1	1
caesar	2		caesar	2	2
caesar	2		did	1	1
did	1		enact	1	1
enact	1		hath	2	1
hath	1		I	1	2
I	1		i'	1	1
I	1		it	2	1
i'	1		julius	1	1
it	2		killed	1	2
julius	1		let	2	1
killed	1		me	1	1
killed	1		noble	2	1
let	2		so	2	1
me	1		the	1	1
noble	2		the	2	1
so	2		told	2	1
the	1		you	2	1
the	2		was	1	1
told	2		was	2	1
you	2		with	2	1
was	1				
was	2				
with	2				

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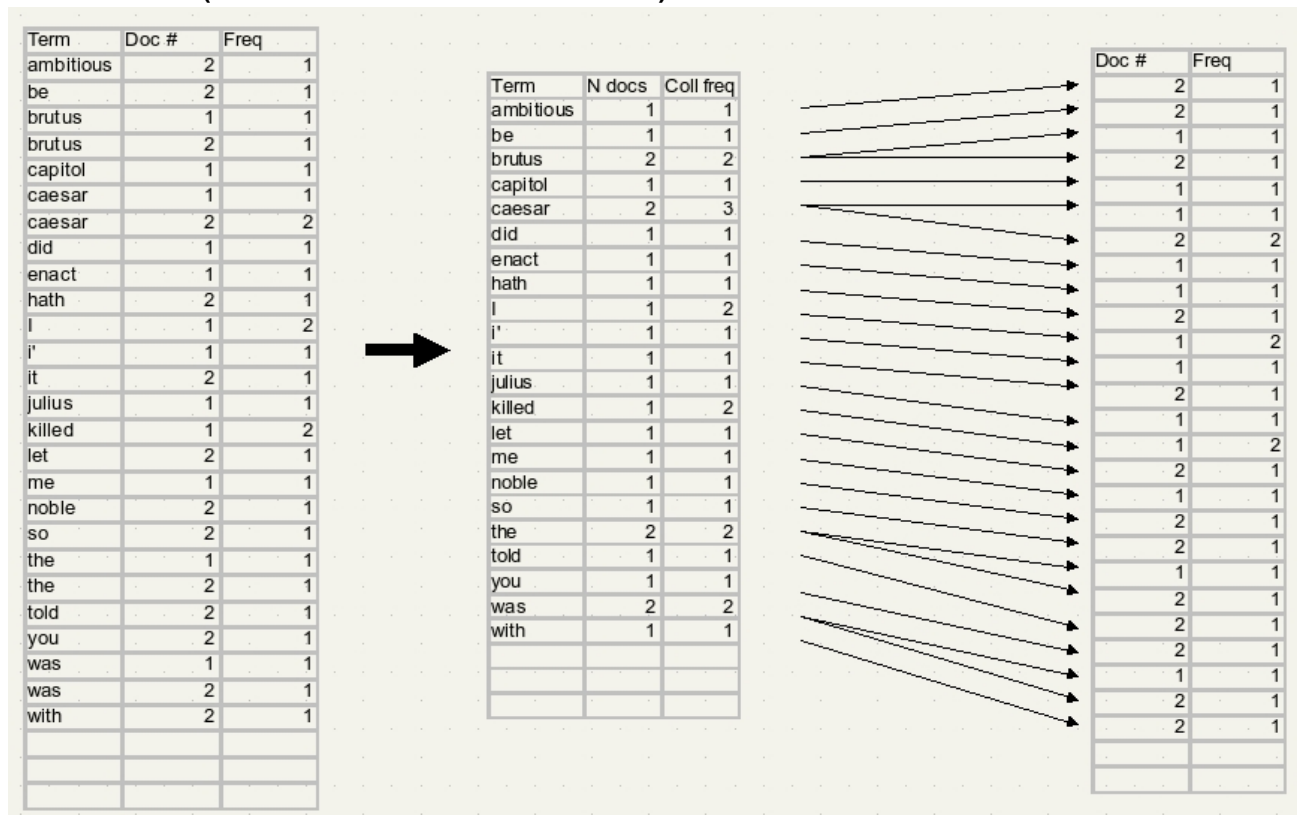
The diagram illustrates the process of creating an index by converting document tokens into a structured table and then merging them into a term-frequency table. A large black arrow points from the left table to the right table.

Term	Doc #
ambitious	2
be	2
brutus	1
brutus	2
capitol	1
caesar	1
caesar	2
caesar	2
did	1
enact	1
hath	1
I	1
I	1
i'	1
it	2
julius	1
killed	1
killed	1
let	2
me	1
noble	2
so	2
the	1
the	2
told	2
you	2
was	1
was	2
with	2

Term	Doc #	Term freq
ambitious	2	1
be	2	1
brutus	1	1
brutus	2	1
capitol	1	1
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caesar	2	2
did	1	1
enact	1	1
hath	2	1
I	1	2
i'	1	1
it	2	1
julius	1	1
killed	1	2
let	2	1
me	1	1
noble	2	1
so	2	1
the	1	1
the	2	1
told	2	1
you	2	1
was	1	1
was	2	1
with	2	1

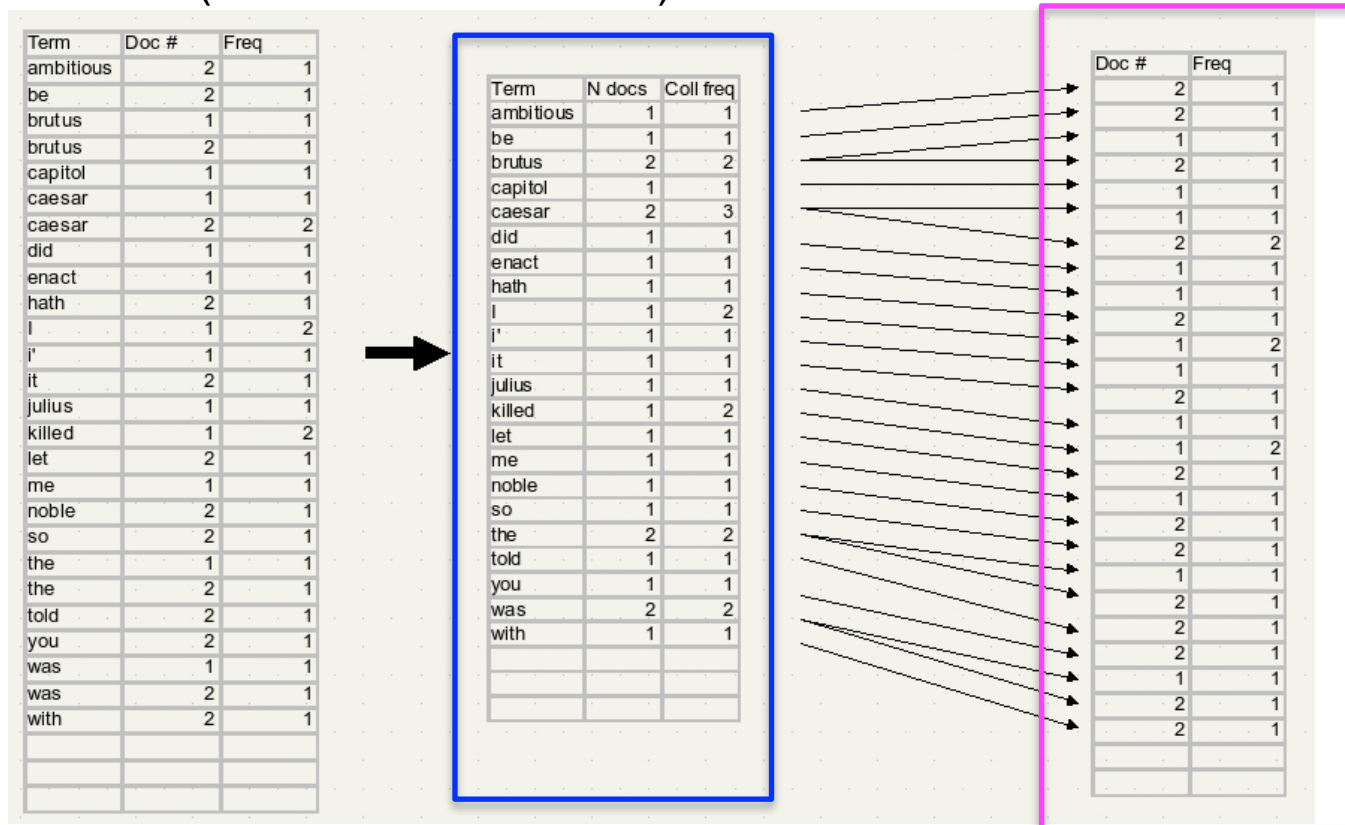
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- Split into dictionary and postings file
- Index allows efficient access to terms, and calculation of statistics about terms and documents (used in retrieval later)



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